TECHNICAL VALIDATION REPORT

Mid-Surface Extraction and Shell Meshing of an Automotive Front Floor Panel (Sonata)

CAE pre-processing in ANSA (BETA CAE Systems) | BIW sheet-metal component | Quality-criteria compliance study

Item	Detail
Component	Front floor panel, Hyundai Sonata (stamped sheet metal, single gauge)
Input geometry	Floor_Panel_stp (STEP, imported to ANSA); working database Floor_Panel_stp.ansa
Tool	ANSA pre-processor (geometry checks, mid-surface, Quality Monitor, mesh tools)
Analysis intent	Shell model for BIW structural / crash / NVH use (solver not specified in the brief)
Mid-surface method	Constant-thickness offset, t = 1 mm
Mesh	45,436 shell elements: 43,824 quads, 1,612 trias; target element length 5 mm (units: mm)
Document purpose	Evidence of method, parameters and quality-criteria compliance for review and validation

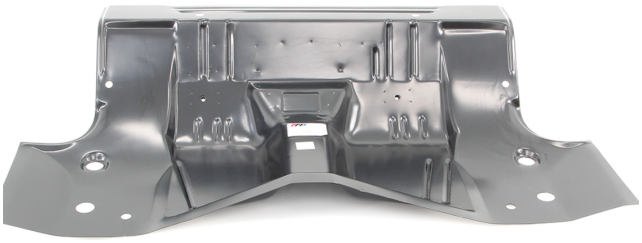


Fig. 1. Reference component: stamped sheet-metal front floor panel

1. Executive Summary

A raw CAD model of a front floor panel was converted into an analysis-ready shell mesh. The geometry was checked and repaired, the mid-surface was extracted with a constant 1 mm offset, the surface was defeatured and rebuilt for clean mesh flow, and a quad-dominant mesh was generated at a 5 mm target length against an eleven-item quality specification.

The final model contains 45,436 shell elements, of which 96.45% are quadrilaterals and 3.55% are triangles, against an allowed maximum of 10% triangles. Mesh-flow rules (no trias on edges or features, no back-to-back or opposite trias, even node counts on holes, no rotational quads, no pattern trias) were applied throughout. Section 8 lists each criterion with its verification method; measured extremes should be copied from the ANSA quality report into the highlighted column before submission.

2. Scope, Inputs and Assumptions

2.1 Scope

- Geometry import and topology repair of the supplied STEP model.
- Mid-surface extraction and assignment of shell thickness.
- Quad-dominant 2D meshing at 5 mm target length to the specified quality criteria.
- Verification of quality and mesh-flow criteria, and documentation of the method.

2.2 Out of scope

- Material assignment, connections (spot welds, adhesive, bolts), boundary conditions, loads and solver run.
- Correlation with physical test data.

2.3 Assumptions

- The panel is a single-gauge stamping with uniform thickness of 1 mm, taken from the brief.
- Model units are millimetres; "5 units" in the brief is therefore 5 mm.
- Holes and openings are retained; small surface splits on flat regions were treated as defeaturing candidates.

3. Quality Specification (Acceptance Criteria)

The following criteria were issued with the task. Every criterion is a pass/fail gate for each element.

#	Criterion	Requirement	Applies to
1	Aspect ratio	5 (max)	All shells
2	Skewness	45 (max)	All shells
3	Warping	15 (max)	Quads
4	Taper	0.5 (max)	Quads
5	Minimum element length	2 mm	All shells
6	Maximum element length	8 mm	All shells
7	Minimum angle, quad	45 deg	Quads
8	Maximum angle, quad	135 deg	Quads
9	Minimum angle, tria	20 deg	Trias
10	Maximum angle, tria	120 deg	Trias
11	Tria percentage	10% (max)	Whole mesh
-	Target element length	5 mm	Whole mesh

3.1 Mesh-flow rules

- No tria elements on edges (free edges, shared edges, non-manifold edges).
- No opposite trias, and no trias touching each other (no back-to-back trias).
- No trias on features: embosses, fillets, holes, logos, dog-houses, welding cylinders, ribs, mounting clips.
- An even number of nodes on every hole or opening.
- No rotational quads and no pattern trias.

4. Methodology

The work followed a fixed sequence in which each stage had an exit check. A stage was not closed until its check passed.

Stage	Activity	ANSA tools / actions	Exit check
1	Import and topology check	STEP import; Geometry checks	Error list reviewed; all auto-fixable errors cleared; manual items assigned
2	Topology repair	Face defeature, Reconstruct, Cons realign / connect, merge / delete	Re-run geometry check: no cracks, needle faces or single Cons
3	Mid-surface extraction	Constant-thickness offset, t = 1 mm	Single surface set, centred between faces, thickness recorded
4	Surface defeaturing	Split, Improve, Reconstruct, Reshape (from shell elements), Join, Length From Perimeter	Continuous crack-free surface; patches meshable as 3/4-sided regions
5	Meshing	Quad-dominant shell mesh, 5 mm target, live Quality Monitor	Counts and criteria within limits region by region

Stage	Activity	ANSA tools / actions	Exit check
6	Failed-element loop	Save Failed (quality panel); Tools > Mask > Elems > Retrieve Failed; Mask Reverse All; Geom > Temp Nodes on visible elements	No remaining failed elements
7	Mesh-flow audit	Visual review of ribs, holes, fillets, tunnel; opposite / back-to-back tria search	Rules in Section 3.1 satisfied
8	Final verification	Quality Monitor statistics; tria percentage	Section 8 compliance table complete

5. Geometry Validation and Repair

The geometry checker was run on the imported model before any meshing work. Its output is shown in Fig. 2 and summarised below; it lists errors by message code, entity and ID.

Code	Meaning	Count seen	Entity / IDs (as listed)	Auto fix	Resolution
E4068	Unchecked faces	7	-	Yes	Full check pass, faces validated
E4071	Needle face	4	FACE 1981, 1987, 2107, 2123	Yes	Defeature and Reconstruct
E4073	Crack between Cons	3	Cons groups beginning 651, 1012, 1384	Yes	Realign / connect boundary curves
E4074	Single Cons (free edge)	20+ (list continues)	CONS 650, 651, 653, 658, 659, 1028-1034, 1036-1040 and others	No	Manual: retrieve, merge or delete, re-check

E4074 items are not auto-fixable, which is why they were handled by hand. Left unresolved, single Cons become free edges in the mesh and break load continuity; needle faces force elements that violate aspect ratio and skewness limits regardless of mesh settings.



Fig. 2. Geometry checker output on the imported model (partial list)

6. Mid-Surface Extraction

6.1 Method selected

Constant-thickness offset, t = 1 mm. A single mid-surface is generated equidistant from both faces of the sheet, and a uniform shell thickness of 1 mm is assigned.

6.2 Rationale and comparison

Aspect	Constant-thickness offset (used)	Variable-thickness / geometry-based
Suitable part type	Uniform-gauge stampings	Castings, moulded parts, multi-gauge parts
Thickness input	One value (1 mm)	Measured per region from the solid

Aspect	Constant-thickness offset (used)	Variable-thickness / geometry-based
Risk on this part	Low: gauge is uniform	Surface deviation and patch fragmentation with no accuracy benefit
Effort	Low	Higher: per-region thickness mapping and clean-up
Effect on bending stiffness	Mesh centred on the neutral axis	Equivalent if thickness is uniform

The mid-surface sits at the centre of the sheet, so shell bending stiffness (which depends on distance from the neutral axis and on thickness) is represented consistently. An off-centre surface would bias bending response in crash and NVH simulations.

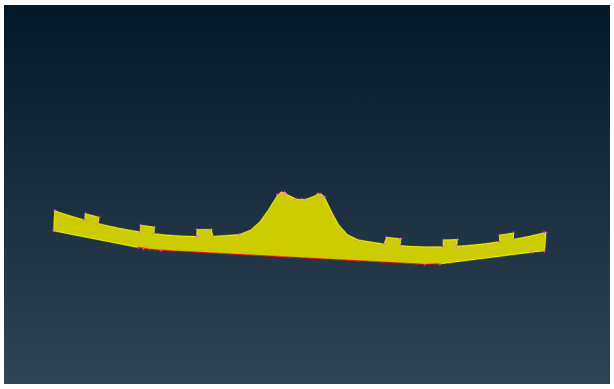


Fig. 3a. Extracted mid-surface, side view

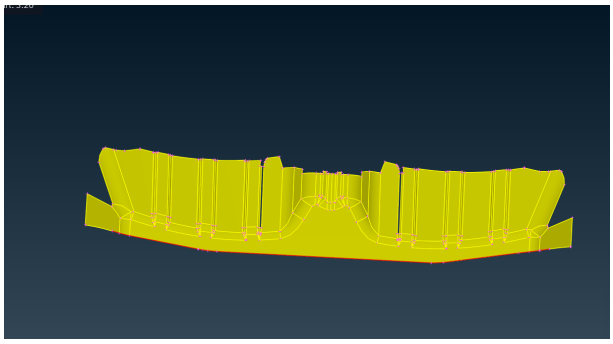


Fig. 3b. Extracted mid-surface, angled view

7. Surface Defeaturing and Meshing

7.1 Defeaturing

Offsetting each face independently leaves overlapping or redundant patches at fillets, ribs and flange transitions. The surface was rebuilt into clean meshable patches:

Tool	Purpose	Where used
Split	Break merged or multi-edged surfaces into simple 3/4-sided patches	Rib walls, tunnel, bead lines
Improve / Reconstruct	Rebuild distorted or self-intersecting patches	Fillet and rib transitions
Reshape (from shell elements)	Regenerate a clean surface from trial-mesh topology	Stubborn regions where CAD patches resisted repair
Join (defeaturing)	Merge unnecessary splits so no artificial patch boundaries remain	Large flat regions
Length From Perimeter	Derive local element size from patch boundary length	Features of very different scale

7.2 Meshing parameters

Parameter	Value
Element type	Quad-dominant shell (CQUAD-type quads, trias only where needed)
Target element length	5 mm
Allowed length range	2 mm to 8 mm
Shell thickness	1 mm, constant
Maximum tria content	10% of elements
Quality control	ANSA Quality Monitor, live during meshing

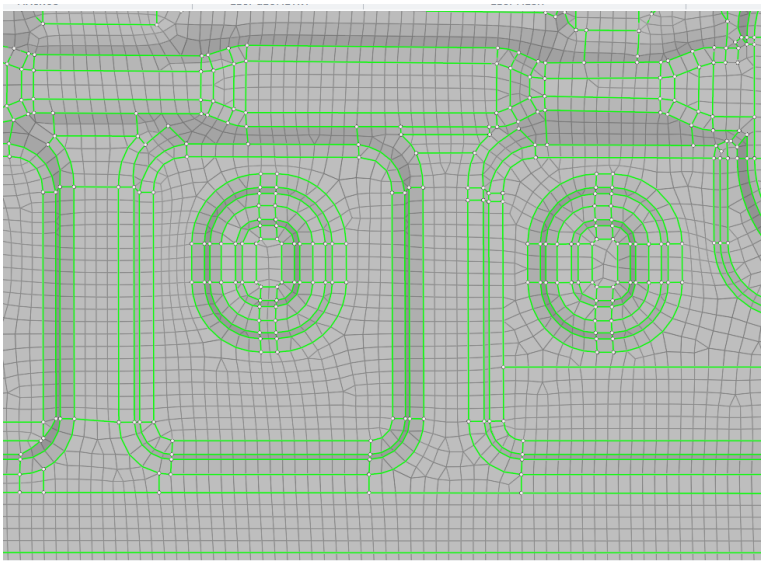


Fig. 4. Concentric washer-ring meshing around fastener bosses, with quad flow into adjoining ribs

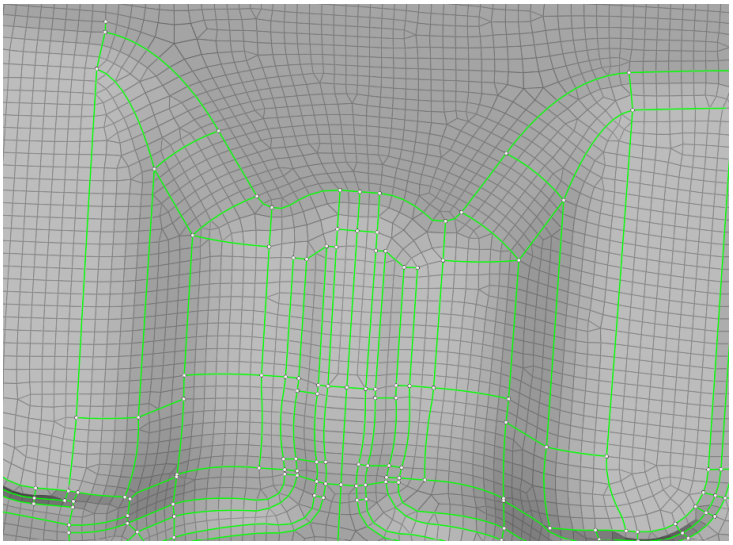


Fig. 5. Mesh transition across the raised central tunnel, no trias on feature edges

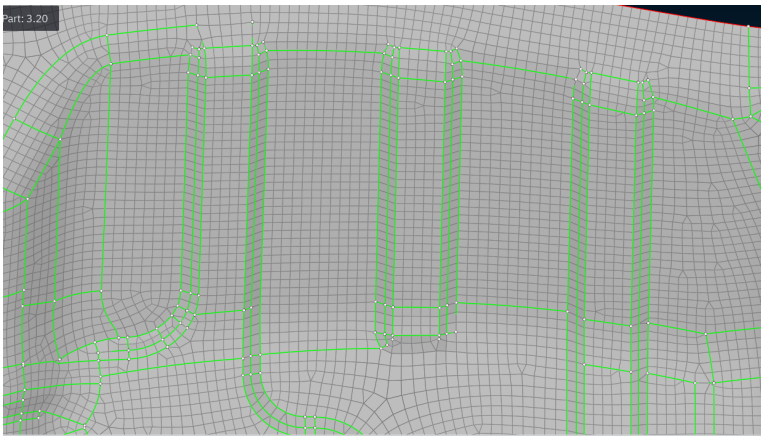


Fig. 6. Row-wise quad flow across the stiffening-rib bank

8. Compliance Matrix

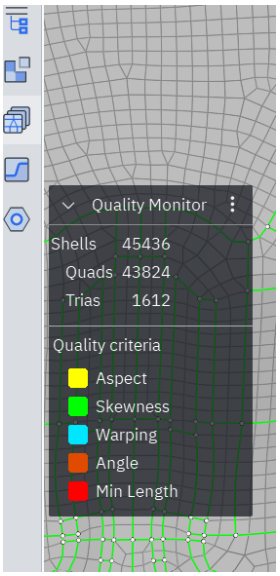
Each criterion is listed with the method used to verify it. The highlighted column is for the extreme value reported by the ANSA quality check on the final database, so that a reviewer can see the margin to the limit.

#	Criterion	Limit	Verification method	Measured (fill from ANSA)
1	Aspect ratio	5 max	Quality Monitor (Aspect), failed-element retrieve	
2	Skewness	45 max	Quality Monitor (Skewness)	
3	Warping	15 max	Quality Monitor (Warping)	
4	Taper	0.5 max	Mesh quality check, Taper	
5	Min length	2 mm	Quality Monitor (Min Length)	
6	Max length	8 mm	Mesh quality check, Max Length	
7-8	Quad angles	45 - 135 deg	Quality Monitor (Angle)	
9-10	Tria angles	20 - 120 deg	Quality Monitor (Angle)	
11	Tria percentage	10% max	Element counts: 1,612 / 45,436 = 3.55%	3.55%

8.1 Mesh-flow checklist

Rule	Method of check	Evidence
No trias on free / shared / non-manifold edges	Visual edge review; free-edge display	Figs. 4-6, 8
No opposite or back-to-back trias	Local review at every tria; Retrieve Failed loop	Figs. 5, 6, 8
No trias on embosses, fillets, holes, ribs, clips	Visual review of every feature	Figs. 4, 6
Even node count on holes	Node count per hole boundary	Figs. 4, 8
No rotational quads, no pattern trias	Visual review of regular regions	Figs. 4, 6, 8

9. Results



Parameter	Value
Total shell elements	45,436
Quad elements	43,824 (96.45%)
Tria elements	1,612 (3.55%)
Tria limit	10%
Margin to tria limit	6.45 percentage points
Shell thickness	1 mm
Target element length	5 mm

Fig. 7. ANSA Quality Monitor panel (left) and summary statistics (right)

The mesh is dominated by quadrilaterals, which represent shell bending stiffness more accurately than triangles and give a more regular, predictable structural response.

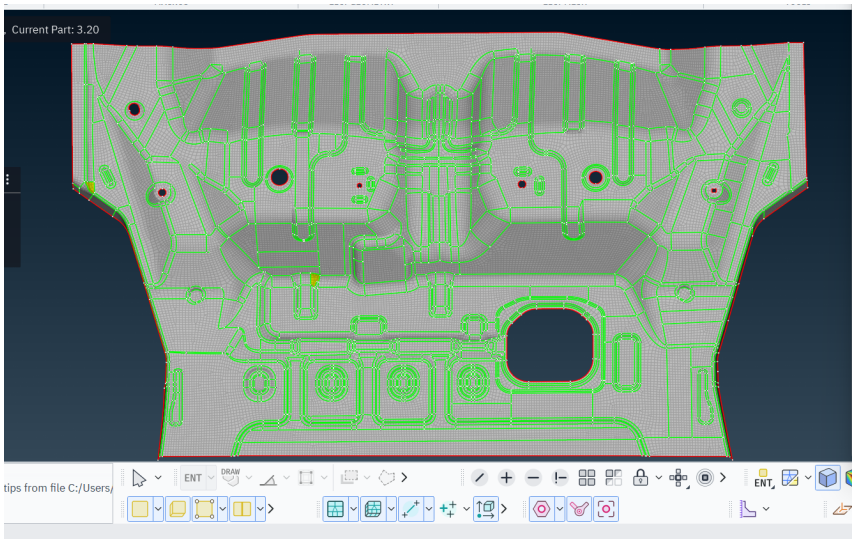


Fig. 8. Complete quad-dominant shell mesh of the front floor panel

10. Risks, Limitations and Recommendations

- **Single Cons and cracks.** Any new CAD revision must be re-checked; these errors reappear on re-import and are the main source of rework.
- **Constant thickness.** Valid for this part. If a revision introduces patches, doublers or tailor-rolled blanks, the mid-surface method and property assignment must be revisited.
- **Element size.** A 5 mm mesh resolves the main ribs and holes but not small beads; crash-critical fine features may need local refinement.
- **Not yet modelled.** Material, connections and boundary conditions are outside this deliverable; the mesh should be re-checked after connections are added, because weld and bolt patterns can force local mesh changes.
- **Reproducibility.** Keep the .ansa database, the checker output and the quality report together as the audit trail.

11. Deliverables Checklist

Item	Status
Native ANSA database (.ansa) with final mesh and thickness	Provided
Technical report (this document)	Provided
Method statement for mid-surface extraction with reason	Section 6
Quality-criteria compliance matrix	Section 8 (measured values to be filled)
Mesh-flow evidence (screenshots)	Figs. 4-6, 8
Any additional formats requested by the reviewer	As per the task instructions

12. Conclusion

The floor panel was converted from raw CAD to an analysis-ready shell mesh through a controlled sequence of geometry validation, constant-thickness mid-surface extraction, defeaturing and quality-gated meshing. The model has 45,436 shell elements with 3.55% triangles against a 10% limit, applies the specified mesh-flow rules, and is documented so that each decision and check can be reviewed independently.